

Lesson 5.17: Make

Unit: Unit 5, Make It, Mend It, Wear It (Apparel and Textile Production and Design) **Topic:** 5.4 Upcycle Challenge project (days 3 and 4 of 6) **Grade:** 6 to 8 (written at grade 7; see Differentiation for the grade 6 support and grade 8 stretch) **Days:** 2 class periods of 40 minutes (days 138 and 139) **Room:** FACS lab, three work stations of about seven students. Needles, pins, and shears are out both days, so the count-out and count-in routine runs both days.

Standards

- NYS module line: NYS ATPD 2. Production c) Demonstrate basic hand and machine sewing techniques by creating or repairing a textile product
- NYS module line: NYS ATPD 2. Production b) Demonstrate safe use and storage of equipment used in hand and machine sewing
- NYS module line: NYS ATPD 2. Production d) Develop sustainability practices for repurposing apparel items
- CTE theme line: CTE HSW 4. Classroom and Workplace Safety c) Apply safety practices when using tools, machines, and equipment
- CTE theme line: CTE HSW 4. Classroom and Workplace Safety e) Participate in housekeeping procedures that help to establish and maintain safe working conditions in the classroom or workplace
- CTE theme line: CTE PSI 3. Troubleshooting Process (Reactive) a) Implement a formal troubleshooting process to solve a given problem by a. Defining the problem being addressed; c. Testing and evaluating to isolate the problem; d. Correcting the problem by implementing changes or repairs
- NYS FACS Learning Standard 2 (Intermediate), Key Idea: Students will know the basic principles of home and community safety. They can demonstrate the skills necessary to maintain their homes and workplaces in a safe and comfortable condition. They can provide a safe and nurturing environment for themselves and others.; Performance Indicator: Students use age-appropriate techniques to select and maintain clothing.
- CDOS Standard 2: Students will demonstrate how academic knowledge and skills are applied in the workplace and other settings.
- CDOS Standard 3a Universal Foundation Skills: Personal Qualities
- National FCS 3.0: 16.4.1 Work skillfully with traditional and newer equipment, tools, and supplies to construct, alter, repair, and recycle textiles and apparel.
- National FCS 3.0: 16.4.5 Show the basic skills for making, altering, repairing, and recycling textiles and apparel.

Enduring understanding and essential question

Big idea: A plan becomes an object through a hundred small decisions made with a needle in your hand. Getting stuck is part of the work, and there is a procedure for it. **Essential question:** When something goes wrong at my station, what do I do before I raise my hand?

Objectives

By the end of this lesson, students will be able to:

1. (Low) Execute the count-out and count-in routine and name the three safety rules for needles, pins, and shears without being prompted.
2. (Mid) Construct their planned item using at least two of the taught joins (hand stitches or a machine seam), following their own order of work, and apply the three-step stuck protocol before asking for help.
3. (High) Evaluate their own item against the finishing checklist and correct at least one problem they found themselves.

Vocabulary

Tier 2 (general academic): milestone, procedure, troubleshoot, correct, evaluate Tier 3 (FACS): knot off, backstitch, whipstitch, running stitch, seam allowance, raw edge, casing, tension, bar tack, finishing. See Vocabulary 05.md for Spanish.

Materials and setup

- Every student's labeled bag: approved plan sheet with the teacher's initials, design brief, material, mock-up
- Three station kits, built to the spec in Lesson 5.5 and unchanged all unit: per station, hand needles counted (ten in the cushion, assorted sizes), thirty straight pins counted, two pairs of fabric shears, three small snips, two seam rippers, four thimbles, four needle threaders, two measuring tapes, two pin cushions, all-purpose polyester thread in four colors, and a small scrap basket
- The station count sheet from Lesson 5.5 (Handout 05.05 - Tool Kit Guide and Count Sheet.md), one per station, with today's row ready
- Self-threading (calyx eye) needles and large-eye embroidery needles, at least three per station, for the reduced-dexterity alternative
- Binder clips and paper clips, at least ten per station, as the pin alternative
- Fabric glue and iron-on hem tape at the teacher's table, for the reduced-dexterity alternative only, plus an iron used by the teacher only
- Stitch cards: the four taught hand stitches and the machine seam, one card each, with a stitched sample and three numbered steps. These are the Topic 5.2 cards; they stay at the station all four project days.
- The "Need help" card, one per station, standing up on a small block so it is visible from the front of the room
- Handout 05.17 - Build Milestones and Stuck Protocol.md, one per student, in the bag from day one of the build
- Cord, shoelaces, or twill tape for drawstrings; buttons in a sorted tin; uncooked rice in a sealed container for the hand warmer path; dried beans or flax as the rice alternative
- Slides: Slides 05.17 - Make
- Setup notes: count the needles and pins in each station kit before the bell and confirm both numbers on the station count sheet. Put the finishing checklist on the wall at each

station, large. Have the three reduced-scope versions ready to name out loud. Keep the seam rippers out where students can reach them, because a student who has to ask for one will pull stitches out with shears instead.

The count-out and count-in routine (both days, every day tools are out)

The six steps, exactly as taught in Lesson 5.5, on the station count sheet. A new Station Manager each sewing day.

1. The Station Manager opens the kit and counts out loud: needles, then pins, touching each one.
2. The Station Manager writes both numbers in today's row on the station count sheet.
3. The station works. Any needle or pin not in a hand is in the pin cushion or the kit.
4. At the five-minute warning everything stops and every needle and pin goes back in the cushion.
5. The Station Manager counts again, out loud, and writes the second number.
6. The numbers match and the teacher initials the row, or they do not match and nobody leaves that station until it is found. If it is not found, the station stands back and the teacher sweeps with the magnet.

Shears are counted too, two per station, six in the room.

During the period. Every needle and pin is either in a hand, in the fabric, or in the pin cushion. Nothing goes on the table, on a chair, in a pocket, on the floor, or in a mouth. A dropped needle or pin is announced out loud and picked up by the person who dropped it, immediately, before they do anything else. Shears are passed handle first with the blades closed. Nobody walks with a threaded needle in their hand; you put it in the fabric first.

Count in, at the five-minute warning. Every needle and pin goes back in the cushion, the Station Manager counts out loud, and both numbers must match today's row. Shears back in the kit, blades closed. No station is dismissed until all three stations match. If a count is short, the whole class stops and we sweep and look until it is found. That rule has never once been worth breaking.

Pre-assessment

The approved plan sheets sorted at the end of Lesson 5.16 are the pre-assessment: who is approved, who is on a simpler path, and who was worried about what. Read the "the part I am worried about is ___" line on all of them before Day 1 and write the three most common worries on the board. Those three get a 90-second demonstration in Step 1 instead of a generic reminder.

Do now and hook (Time: 5 min Day 1, 4 min Day 2)

Both days the do now runs while the Station Managers run the count-out, so the tools are counted before a single pin moves.

Day 1, slide 1. On the board: “Write today’s milestone in your own words, then the first three things you will do in order.”

Day 1 hook: state the milestone out loud and post it. By the end of today, my piece is cut, marked, pinned, and at least one full seam or edge is stitched. “That is the bar. If you hit it, tomorrow is finishing. If you do not, tomorrow is catching up, and I would rather you hit it.”

Day 2, slide 9. On the board: “Write the two things left to do and which one you do first.”

Day 2 hook: “Today it becomes a thing somebody can use. At the end of the period it either works or it does not, and the finishing checklist decides.”

Procedure

Day 1 of the build: cut, pin, and the first seam

Step	Teacher will	Students will	Time
Do now, count-out, and hook	Slide 1. Run the Day 1 do now and hook above while the Station Managers count out.	Count out (Station Managers). Write the milestone and the first three steps.	5 min
1. The stuck protocol and the day’s three demonstrations	Slides 2 to 5. Teach the stuck protocol as a procedure and post it at every station: (1) read the step on your plan sheet again and put your finger on where you are; (2) look at the stitch card for that stitch and do two practice stitches on a scrap; (3) ask your station partner and show them the exact spot. Then, and only then, stand the “Need help” card up and keep working on a part you can do while you wait. Say the reason plainly: “Three of you get stuck on the same	Read the protocol out loud as a class once. Watch the three demonstrations.	5 min

	thing at the same minute. The protocol means two of you are moving again before I get there.” Then give the three 30-second demonstrations for the worries from the plan sheets (most commonly: starting a knot that holds, turning a corner, and how far from the edge to stitch).		
2. Build	Slide 6 (the milestone and the clock). Circulate on a fixed loop so every station sees you every three minutes. Answer only the “Need help” cards, in the order they went up. Keep the fabric shears moving; check the marked line before you let anyone cut. At minute 12 say “everyone should be cut and marked.” At minute 20 say “everyone should be pinned.”	Cut on the marked line, mark the stitch line, pin or clip, and stitch the first full seam or edge. Use the stuck protocol. Keep the plan sheet open on the table.	22 min
3. Milestone check and clean	Slide 7. Walk the three stations with the clipboard and tick each student against the Day 1 milestone: cut, marked, pinned, one seam or edge stitched. Name out loud, without	Self-check the milestone on the handout. Clean the station. Bag and label everything, including loose pieces.	5 min

	naming students, how many hit it. Assign the catch-up or the extension for tomorrow to each student in one sentence. Run the housekeeping routine: scraps in the basket, thread ends off the floor, tools to the kit, bags labeled and stacked.		
Closure and count-in	Slide 8. Count-in with the Station Managers, all three counts confirmed out loud, then dismissal.	Count in. Write the one-line closure below.	3 min

Day 1 total: 5 plus 5 plus 22 plus 5 plus 3 = 40 minutes.

Day 2 of the build: finish it

Step	Teacher will	Students will	Time
Do now, count-out, and hook	Slide 9. Run the Day 2 do now and hook above while the Station Managers count out.	Count out. Write the two remaining tasks in order.	4 min
1. The finishing checklist, read out loud	Slide 10. Read all seven lines of the finishing checklist and post it at every station: every seam is stitched closed with no gap wider than the head of a pin; every thread end is knotted and trimmed to a quarter inch; the closure works (the drawstring pulls, the button fastens through the hole,	Read the checklist. Mark which lines they already pass.	4 min

	the flap folds); no pins are left in the item; the item does the job it was made for, tested with the actual contents; the raw edges are folded, whipstitched, or otherwise dealt with; and your name is on the bag, not sewn into the item. Then name the two paths: finished early means the second-item or station-helper option, behind means the reduced-scope version on your plan sheet, and neither one costs points.		
2. Build and finish	Slide 11 (the checklist and the clock). Same fixed loop. Same “Need help” order. At minute 10 say “if you are behind, switch to your reduced-scope version now, not at minute 20.” At minute 18 say “start finishing: knots, trims, closure.” Hand out the early-finisher options as students hit the checklist.	Finish the item. Test the closure with real contents. Run the checklist line by line and fix what fails. Early finishers start a second small item or move to the helper role.	22 min
3. Finishing checklist self-check and the handoff to the gallery	Slide 12. Circulate with the clipboard and confirm the seven lines on each	Self-check all seven lines and initial. Fix one problem found by the check, not by	5 min

	finished item, initialing the handout. For anything unfinished, write on the handout what remains and schedule it: the first ten minutes of Lesson 5.18 Day 1 are a finishing window for exactly this. Say what tomorrow is: display card, then the 45-second explanation.	the teacher. Put the finished item, the plan, the brief, and the mock-up in the bag.	
Closure and count-in	Slide 13. Count-in, all three confirmed.	Count in. Write the closure line.	3 min
Buffer	Two minutes are the stated buffer for the count that does not match, the shears that go missing, or the student whose closure fails at minute 20.		2 min

Day 2 total: 4 plus 4 plus 22 plus 5 plus 3 = 38, plus the 2-minute buffer = 40 minutes.

The stuck protocol (posted at every station, restated Day 1 and Day 2)

Before you raise your hand, three things:

1. **Read it again.** Find the step on your plan sheet and put your finger on it. Say out loud where you are.
2. **Look at the card.** The stitch card shows the stitch in three steps with a real sample. Do two stitches on a scrap.
3. **Ask your partner.** Show them the exact spot. Not “it’s not working,” but “this is where the thread keeps pulling through.”

Then stand up the “Need help” card and keep working on something you can do. Cards get answered in the order they go up.

What to do if you finish early

You do not get to sit. Pick one:

1. **A second small item** from the Easy list, from the scrap bin, planned on the back of the plan sheet, approved in thirty seconds. Most students who finish early pick the headband, the phone sleeve, or a patchwork square.
2. **Station helper.** Thread needles for other students, hold fabric while a partner stitches a corner, demonstrate a stitch from the stitch card, or run the scrap basket. Helping is scored in the participation and safety category, and a helper who teaches a stitch correctly is noted on the rubric's notes line as evidence for criterion 4.
3. **Make it better.** Add a reinforcement at a stress point (a bar tack at each end of the drawstring channel or at the corners of a pocket), a second row of stitching on the seam that carries the weight, or a hand-stitched detail. Say which one you added on the display card.

What to do if you fall behind

The reduced-scope version is already written on your plan sheet from Lesson 5.15 and 5.16. You do not invent it under pressure. The most common reductions, printed on Handout 05.17:

Original	Reduced scope
Jeans-leg pencil case with a flap and a button	Jeans-leg open-top pouch, one seam across the bottom, whipstitched edge, no flap
Towel produce bag with a drawstring	Towel flat pouch, two backstitched sides, folded top, no casing
Shirt pillow cover, three seams	One patchwork square using the shirt fabric, four sides joined
Pillowcase drawstring bag	Pillowcase flat bag, one seam, a ribbon tied at the top instead of a casing
Sweatshirt cuff wristlet with an inside pocket	Sweatshirt cuff wristlet, no pocket, both raw edges whipstitched
Patchwork square of six pieces	Patchwork square of two pieces, one seam, edges whipstitched

Switching to the reduced scope is a decision, not a failure, and the rubric scores what you turned in against the same five criteria. Switching at minute 10 produces a finished object. Switching at minute 35 does not, and that is the only reason the clock calls are on the slides.

Questions to ask

Monitor understanding:

- Which step of your order of work are you on? Point at it.
- Which stitch is that, and why that one here?

- How far from the edge are you stitching? Is that what your plan said?
- You just knotted off. Did you trim it, and how long is the tail?
- What did you try before the card went up?

Deepen learning (why and how, tied to their lives):

- Your seam opened when you pulled it. What does that tell you about stitch length, and what would a factory do differently?
- You are about to whipstitch a raw edge. What happens in three months if you skip it?
- Somebody in this room is doing the same stitch faster than you. Watch them for ten seconds and tell me the one thing they do differently.
- The item works but it does not look finished. Name the difference in one sentence.

Check for understanding

Day 1 clipboard: each student ticked against the four-part milestone (cut, marked, pinned, one seam or edge stitched) plus a yes or no on “used the stuck protocol before the card went up,” which the station partners will tell you honestly if you ask them and not the student. A student missing two of four milestone parts is named for the reduced scope tomorrow, in one sentence, at the end of Day 1 and not on Day 2.

Day 2 clipboard: the seven-line finishing checklist confirmed on each item, initialed on the handout. A student who found and fixed a failure themselves gets a mark that feeds Rubric 05 criterion 2 at the Exceeds level. Tool safety and the station routine are ticked both days and feed criterion 4, which is scored across all four project days, not just today.

Closure (Time: 3 min each day, count-in first)

Day 1, one line: “I hit or missed the milestone because . ***Tomorrow I start with .***” Two students read.

Day 2, one line: “The hardest stitch on my item was . ***The part I am proudest of is .***” Three students read. These lines are the first draft of the display card in Lesson 5.18, so tell students to keep the handout in the bag.

Differentiation and supports

- ELL: the stitch cards are three numbered pictures plus a stitched sample, so a stitch can be learned without reading; the stuck protocol is posted as three icons (an eye on the plan, a hand on the card, two people talking); the finishing checklist is seven checkboxes with a picture cue each; the Spanish words for needle, thread, knot, stitch, cut, and finished are on the handout and in Vocabulary 05.md; a bilingual station partner where possible.
- IEP and 504 (general): pre-threaded needles kept in a labeled card at the teacher’s table, at least six ready at all times; the reduced-scope version already written so no decision is made under pressure; the option to work at a single station all period with no rotation; extended time in the Lesson 5.18 Day 1 finishing window; hand-over-hand instruction for the first two stitches of any new technique; a chair with a table at a

height that lets the elbow rest, because most dropped needles come from a bad working height.

- Grade 6 support: the Easy paths only, the material pre-marked by the teacher, and a milestone of “cut, marked, pinned, and three inches of one seam stitched” on Day 1.
- Grade 8 stretch: add one reinforcement at a stress point and be able to say why that spot carries the load; use a third stitch beyond the two required; or take the station-helper role and teach a stitch from the card to two other students, which is noted on the rubric’s notes line as evidence for criterion 4.
- UDL checkpoint used: multiple means of action and expression (the item can be built with needle and thread, with clips instead of pins, or with the teacher’s hands guiding the student’s, and the product is scored the same) and multiple means of engagement (a posted milestone, a visible clock, and a procedure for getting unstuck lower the cost of struggling in public).
- No-machine alternative: every path can be built by hand, so a room without machines runs both days as written; where machines exist, they are open under the Lesson 5.10 rules and a machine seam counts as a taught join. **Reduced-dexterity alternative, required on both of these days:** a self-threading (calyx eye) needle or a large-eye embroidery needle with heavier thread; a needle threader; binder clips instead of straight pins; a hoop, clamp, or a piece of tape to hold the fabric still so only one hand has to work; felt instead of woven fabric, because felt does not fray and needs no edge finishing; iron-on hem tape or fabric glue for one of the two required joins, with the second join hand stitched so the stitch objective is still met; and a longer running stitch on a larger-scale item, which is easier to control than small stitches on a small item. A student using these alternatives is scored on the same rubric with no note of the accommodation on the score, only on the notes line if it changes what was made.

Assessment

Objective	Standards met	Learning task	Assessment (formative or summative; tool)
1. Run the count-out and count-in and name the tool rules	ATPD 2. b); HSW 4. c), 4. e); CDOS 3a Personal Qualities; National FCS 16.4.1	Count-out and count-in, both days; housekeeping in Step 3	Formative; station count sheet numbers matching with the teacher’s initials, plus the clipboard tick; feeds Rubric 05 criterion 4 and the participation and safety category
2. Construct the item with two of the taught joins, using the stuck protocol	ATPD 2. c), 2. d); PSI 3. a) a., c., d.; CDOS 2; National FCS 16.4.5	The build, both days Step 2	Formative feeding summative; Day 1 milestone clipboard and Day 2 finishing

			checklist; the item is scored on Rubric 05 criterion 2
3. Evaluate the item against the finishing checklist and correct one problem	ATPD 2. c); PSI 3. a) c., d.	Self-check, Day 2 Step 3	Formative feeding summative; the initialed handout; a self-found and self-fixed failure is Exceeds evidence on criterion 2

The finished item is deliverable 3 of Project 05 - Upcycle Challenge.md, scored on Rubric 05 criteria 2, 3, and 4.

Homework

None. Nothing goes home: the item, the material, the needles, and the pins all stay in the room. A student who wants to keep working takes the plan sheet home and writes, not sews.

Connections

Career and technical connection: this is what an alterations tailor does all day, and Lesson 5.19 makes that explicit. The stuck protocol is a troubleshooting procedure, which is the Problem Solving and Innovation theme's reactive process and the same habit a technician uses. Inside the building: the art teacher for anyone adding a stitched design; the special education teacher or occupational therapist for the reduced-dexterity kit, because they will have better tools than the FACS room does.

Sources

- NYS Middle Level CTE FACS Content Module 03, Apparel and Textile Production and Design (nyctecenter.org, June 2018), lines 2. b), 2. c), 2. d). Copy in 02 Standards/NYS Module PDFs/.
- NYS Middle Level CTE Theme Modules: Health, Safety, and Wellness 4. c) and 4. e); Problem Solving and Innovation 3. a) a., c., d. Same folder.
- NYS Learning Standards for Health, Physical Education, and Family and Consumer Sciences (1996), Standard 2, Intermediate, FACS strand. Wording per 02 Standards/STANDARDS SOURCES - web research.md, section 2.
- CDOS Standard 2 and Standard 3a Personal Qualities, per 02 Standards/STANDARDS CROSSWALK.md, Part A.7.
- National Standards for Family and Consumer Sciences Education 3.0 (LEAD FCS Education, 2018), competencies 16.4.1 and 16.4.5, per 02 Standards/STANDARDS SOURCES - web research.md, section 4.16.
- The four taught stitches, the stitch cards, the station count sheet, and the six-step count-out and count-in routine come from Topic 5.2 of this unit (Lessons 5.5 to 5.10).

Teacher notes

- The fixed circulation loop matters more than anything else on these two days. Three stations, every three minutes, same order. Students stop chasing you when they can predict you.
- The “Need help” card is the single change that makes a 21-student sewing room work. Without it, seven hands are up and nobody is sewing. With it, the student keeps working while they wait.
- Common mistake: knotting off and cutting the thread flush to the fabric, so the knot pops on the first pull. A quarter-inch tail is on the finishing checklist for that reason. Show a popped knot on a scrap; it takes ten seconds and they never forget it.
- Second common mistake: stitching too close to the raw edge, so the seam tears out. The plan sheet’s seam allowance line and the Day 1 clock call at minute 20 both exist to catch this.
- Third common mistake: a student unpicks perfectly good stitching because it is not pretty. Name the difference out loud between a flaw that affects function and one that does not. Pull out the seam ripper only for the first kind.
- Say the reduced-scope call at minute 10 on Day 2, not later. A student who switches at minute 10 goes home with something they made. A student who switches at minute 30 goes home with pieces, and that is the teacher’s mistake, not theirs.
- The count that does not match is the two-minute buffer’s real purpose. It happens. Stop everything, sweep, look under the table, check cuffs and pockets. Finding it takes about ninety seconds and it teaches the whole routine better than any reminder.
- Budget note for the unit plan: thread (three spools of all-purpose polyester, about \$3 each), hand needles (a pack of 25 assorted, about \$4, plus a pack of self-threading needles, about \$5), straight pins (a box, about \$3), needle threaders (two packs of 10, four per station, about \$3 a pack), fabric shears (six pairs, two per station, about \$12 each, a one-time purchase), seam rippers (six, two per station, about \$3 each), thimbles (two packs, twelve in all, four per station, about \$4 a pack), cotton cord for drawstrings (a spool, about \$4), buttons (a jar of assorted, about \$6, or free from a donation drive), fabric glue (one bottle, about \$4), iron-on hem tape (one roll, about \$4), binder clips (a box of 12, about \$2), uncooked rice (a two-pound bag, about \$2). Most of this is a one-time purchase that serves the sewing topic and the project together. Materials are free from the scrap bin and the donation drive.